

EDUCATION

University of Massachusetts Amherst

August 2024 - May 2026

Master's in Computer Science

GPA: 4.0 / 4.0

Coursework: Machine Learning, Applied Information Retrieval, Mobile & Ubiquitous Computing, Optimization in Computer Science, Probability & Statistics

JSS Academy of Technical Education

Aug 2016 - Aug 2020

Bachelor of Engineering in Electronics and Communications Engineering

GPA: 3.8 / 4.0

Coursework: Operating Systems, Python, Information Theory, Communication Networks, Engineering Math, Microprocessors, IoT & Embedded Systems, C++ (OOP)

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, Bash, C#

Data & Systems: SQL data pipelines, structured data storage, performance profiling & optimization, API/systems integration, Docker, Git, Linux/Bash scripting

ML/AI & NLP: PyTorch, TensorFlow, Hugging Face, scikit-learn, RAG, LLM Evaluation, Prompt Engineering, NumPy, pandas

Additional: Jupyter, MATLAB, Edge AI/Embedded Systems (RTOS, Sensor Fusion), Unity/C#

EXPERIENCE

CIIR Lab, UMass Amherst

LLM Research Engineer

Jun 2025 - Present

- Designed and iterated multi-hop prompts integrating evidence across document pairs, improving factual integration scores by 42%.
- Introduced phrasing constraints to reduce hallucinations and yes/no leakage, increasing evaluation pass rate from 58% to 81%.
- Built a five-dimensional evaluation rubric and automated Python-based batch scoring pipeline to assess objectivity, fluency, evidence integration, phrasing, and response structure.
- Streamlined dataset-level evaluation workflows using batched LLM API calls, increasing question-generation throughput by 3x across expanding benchmark sets.

Accenture

Application Development Associate

Dec 2020 - Sep 2022

- Analyzed application performance using SQL queries and SAP transaction monitoring logs to identify top 5 bottlenecks, improving response time by ~30%.
- Led two SAP delivery projects across integrations, client requirements, and cross-functional issue resolution.
- Resolved 15 critical UAT bugs before launch, reducing monthly downtime by ~10 hours.
- Introduced JIRA sprint workflows that reduced delivery time by 15%.

Experts Hub

Software Engineer Intern

Jan 2019 - March 2019

- Developed a Python-SQL data pipeline processing 50K+ records daily with 95% accuracy.
- Built a React Native weather-tracking app used by 100+ users; won "Most Intuitive Design" award.

Digital Shark with Texas Instruments

Software Engineer Intern

Jun 2019 - August 2019

- Collected and processed real-time sensor data (temperature, humidity, rainfall) via IoT hardware deployed across local farms and pastures; built a Python pipeline to parse and store readings in a structured database.
- Generated personalized irrigation and crop recommendations for local small-scale farmers based on real-world field sensor data.

ACADEMIC PROJECTS

Adaptive Retrieval Depth Optimization in RAG | *Python, BM25, Flan-T5, HuggingFace, NumPy*

Jan 2026 - May 2026

- Framed retrieval depth k as a per-query optimization variable in RAG pipelines; designed a confidence-based adaptive stopping rule requiring zero fine-tuning or task-specific supervision.
- Evaluated on Flan-T5 Small/Base across SQuAD/TriviaQA; achieved Pareto dominance on SQuAD over fixed- k baselines.
- Reduced average retrieval depth to 1.19–1.60 passages vs. fixed $k=5$, cutting retrieval cost by up to 68% with no accuracy loss.

Unity game called "Chord Strike" | *Unity, C#, Audio Mixer, Animator FSM*

Sep 2024 - Dec 2024

Built a fantasy-themed rhythm game with real-time pitch detection, procedural enemy AI, and state-machine-driven animations across 3 difficulty levels; implemented custom audio event triggers and scene management in C#.

Gesture-Based UAV Drone Control | *XGBoost, ESP32, MPU9250 IMU, Python*

Sep 2024 - Dec 2024

Achieved 98.8% classification accuracy with XGBoost on 430 IMU samples across 8 gesture classes; outperformed SVM (57%) and KNN (78%), enabling real-time hands-free UAV control via ESP32 + MPU9250 sensor fusion.

Automated Weather Monitoring System for Small-Scale Farmers

Jan 2019 - Feb 2019

Built a small-scale home automation system integrating weather monitoring with other automated household devices, leveraging IoT weather data to generate personalized irrigation and crop insights.

RESEARCH AND PUBLICATIONS

Sunil et al. (incl. **Shuchi Mishra**). "[Memory Poisoning Attack and Defense on Memory Based LLM-Agents.](#)" arXiv:2601.05504 [cs.CR], Jan 2026. - Designed and evaluated adversarial robustness benchmarks across GPT-4o-mini, Gemini-2.0-Flash, and Llama-3.1 on clinical data; engineered composite trust-scoring and memory sanitization defenses for production LLM agents; complements hallucination-reduction and RAG reliability research at CIIR Lab.

Shuchi Mishra et al. "[Traffic Management System using V2V Communication employing Li-Fi and IoT.](#)" *IJSRED*, Vol. 3, Issue 3, May–June 2020.